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The Politics of Climate Change in the Developing World

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Abstract

Climate change politics in the developing world remains understudied, despite the region's acute vulnerability and centrality to climate futures. This article synthesizes emerging research across three domains: public opinion and climate salience, the political effects of climate exposure, and the institutional production of climate risk. We highlight a central paradox: Widespread public concern often exists alongside low climate literacy, suggesting that political salience stems from lived experience with environmental degradation rather than scientific attribution. Yet the literatures on climate and environmental politics have developed along separate tracks, limiting conceptual integration and obscuring how local environmental decline manifests as climate risk. Turning upstream, we examine how institutions shape climate exposure itself. Climate vulnerability, we argue, is not merely inherited but also politically produced and unequally distributed through the institutions that govern carbon sinks, build adaptive capacity, and allocate political voice. We identify critical gaps around the distributive politics of adaptation, representation, and institutional sources of climate exposure.

INTRODUCTION

Climate change is widely recognized as one of the most formidable policy challenges of the twenty-first century. However, the development of the political science scholarship on the topic has long lagged behind that of other disciplines, particularly economics, geography, demography, and the natural sciences. Research on the politics of climate change has been largely led by scholars from other disciplines and published primarily in specialized or interdisciplinary journals, such as *Global Environmental Change* and *Nature Climate Change* (Bernauer 2013). In recent years, this gap has narrowed (Grossman et al. 2026), but the literature remains heavily skewed toward developed countries. Much of the literature continues to focus on the politics of mitigation in high-income countries, including studies of international climate negotiations (Bechtel et al. 2019), the design of emissions trading schemes (Green 2021), and mass attitudes in the United States and Europe (Egan & Mullin 2017). Since the 2015 Paris Agreement, domestic distributive politics has been viewed as central to climate policies and outcomes. However, this national turn remains concentrated in the developed world. This focus is not without justification: Wealthy countries have contributed a disproportionate share of historical greenhouse gas emissions and arguably bear greater responsibility for addressing the climate crisis.[1] Nevertheless, this emphasis overlooks a critical reality: The developing world is where climate change's impacts are most acute (Adom 2024) and where countries are least equipped to address its effects.[2] The political dynamics of climate governance are most urgent and least understood in low- and middle-income countries (LMICs).

This article centers the developing world in the study of climate politics. We focus on three lines of inquiry. First, how politically salient is climate change in developing countries? Are citizens in developing regions aware of the phenomenon? To answer these questions, in the section titled Climate Change Attitudes in the Developing World, we examine available public opinion data that measure levels of climate awareness, climate concern, and mass support for climate policies and survey the literature on mass public opinion across developing countries. In the section titled The Effects of Climate Change Exposure, we explore our second line of inquiry: whether and how exposure to climate-related shocks, such as floods, droughts, or heat waves, influences political attitudes and electoral outcomes. Third, in the section titled The Effects of Institutions on Climate Management, we reverse the causal chain, shifting the focus from the effects of climate exposure to examine its key explanatory causes: political institutions.

We advance four core claims. First, we highlight a paradox: In much of the developing world, low levels of climate literacy coexist with high levels of concern about climate change. This paradox suggests that the political salience of climate change does not require a scientific understanding or attribution to anthropogenic causes. Instead, citizens often experience climate risk through local environmental disruptions, such as erratic rainfall, water scarcity, or crop failure, that are more immediate and observable than global climate patterns. These experiences, although not consistently recognized as climate change, make the phenomenon politically salient.

This observation leads to our second core claim: The study of climate politics and that of environmental politics have evolved mainly along separate tracks. Within comparative politics, the study of environmental politics has traditionally focused on local and national struggles over pollution, land use, biodiversity loss, water access, and conservation. By contrast, climate politics

[1] High-income countries contribute about 37–40% of current annual global CO_2 emissions (as of 2022). By contrast, lower-middle-income countries contribute 10–12%, and low-income countries contribute less than 1%—estimates derived from World Bank data (2022) and the Global Carbon Project (Ritchie 2023).

[2] According to recent estimates, income losses in low-income countries are 60% higher than in high-income countries (Adil et al. 2025).

emerged in the 1990s and 2000s as a transnational, often technocratic area of study within international relations (see Bernauer 2013). This international focus tends to downplay the national and subnational politics of climate governance, especially in the developing world (Bernstein 2001, Dryzek 2022). Similarly, while environmental politics offers tools for understanding how weak enforcement, clientelism, and distributive conflict shape resource use, it rarely links these dynamics to the broader challenges posed by climate change. Case studies of environmental degradation in LMICs, such as deforestation, coral reef loss, or aquifer depletion, are often not framed as contributions to climate politics, even though they represent key mechanisms through which climate vulnerability is realized. In developing regions, we argue, these domains are inseparable. Climate change often becomes politically salient through the politics of local environmental decline. Bridging these literatures is essential for tracing the causal chain from institutions to climate exposure to political response.

Third, we observe that exposure to climate shocks does not automatically translate into political action. We review and classify studies into two distinct channels: an attitudinal channel, which links personal experience to increased concern or salience, and an accountability channel, which examines whether voters reward or punish incumbents for climate-related events. Each channel rests on different assumptions and exhibits distinct methodological challenges. However, both literatures face a deeper theoretical problem: Climate change is not a discrete, observable event. Without prior knowledge or interpretive frames, climate exposure may fail to generate meaningful updates to beliefs or behavior. This helps explain the mixed empirical findings across studies, especially in the developing world, where media access, trust in government, and political efficacy vary widely.

Fourth, we argue that climate vulnerability is not exogenous but is, partially, politically produced. Governments and institutions at different levels of government shape climate vulnerability through carbon sink management (Hochstetler & Keck 2007, Buntaine et al. 2015, Mangonnet et al. 2022, Sanford 2023, Calacino 2025, Xu 2025), the distribution of adaptive capacity (Adger et al. 2009, Eriksen et al. 2020), and decisions about inclusion in climate decision-making (Hochstetler 2020, Slough et al. 2021, Dolšak & Prakash 2022, Baragwanath et al. 2023, Gulzar et al. 2024). However, political science has only begun to document how institutions shape the geography of climate harm.

We observe that the next frontier in climate politics lies in understanding the political economy of adaptation. The dominant framing of climate politics, especially in high-income countries, has centered around mitigation (Dolšak & Prakash 2022). However, for much of the developing world, this focus is misaligned with the lived realities and political imperatives on the ground. The existing literature on adaptation is heavily fragmented, with isolated case studies of particular interventions or anecdotal evidence about local initiatives, and lacks the kind of systematic, comparative analysis that would allow generalizable insights.[3] We observe that where investments in formal adaptation from the state are scarce or absent, households and communities resort to bottom-up adaptation strategies (e.g., migration, private cooling technology, new labor arrangements) (Liu & Xu 2025), thereby dampening demands on the state and reducing the need for climate literacy and accountability. However, we know little about when these bottom-up adaptation responses can substitute or be coproduced with state-led programs.

Taken together, these claims point to a set of distinctive political dynamics that remain underexplored in the study of climate governance. In the developing world, climate politics is often made visible through local environmental decline, shaped by weak or uneven institutional capacity and mediated by patterns of exclusion from climate decision-making. By tracing the causal chain

[3] But see Woodruff & Regan (2019) and Mizuno & Okano (2024).

from climate exposure to political behavior—and reversing it to consider how institutions shape vulnerability itself—this article integrates fragmented literatures and clarifies the mechanisms that underpin the reciprocal relationship between climate change and political outcomes.

CLIMATE CHANGE ATTITUDES IN THE DEVELOPING WORLD

This section reviews public opinion research in the developing world on climate awareness, concern, and policy support. Climate awareness refers to an individual's recognition of climate change—its existence, causes, and impacts—as a global phenomenon partly driven by human activity. Climate concern refers to the extent to which individuals perceive climate change as a serious issue, encompassing both cognitive and emotional responses. Climate policy support denotes public endorsement of government actions to mitigate (e.g., carbon pricing, renewable energy investments) or adapt (e.g., early warning systems, climate-smart agriculture) to climate change.

Understanding mass climate attitudes in developing countries is a first-order concern. In democracies and hybrid regimes, in particular, governments are more likely to act on climate issues if citizens demands action. Individuals who do not perceive climate change as sufficiently urgent or doubt its anthropogenic drivers are less likely to support costly policies for mitigation (Gazmararian & Milner 2025) or collective adaptation efforts (Steg 2023). Without public concern, climate policies may lack salience, and more immediate development priorities will take precedence. Similarly, the ability of climate skeptics to block climate action depends on prevailing public beliefs.

Mapping attitudes in regions with high adaptation needs can help assess whether climate inaction stems from an awareness–action gap: the disconnect between growing knowledge of climate change and limited behavioral changes (Colombo et al. 2023). Without understanding climate attitudes, climate inaction is often explained simply as a lack of public demand. However, low demand may reflect a range of different underlying factors, such as insufficient concern, limited climate knowledge, misattributed causes, low expectations of communal cooperation, low trust in government, low self-efficacy, and weak climate salience, that scholars rarely examine systematically.

Based on the growing body of work and our analysis, we report a set of striking and underappreciated patterns. While awareness of anthropogenic climate change remains uneven, particularly in sub-Saharan Africa, concern about climate impacts is high across the developing world—often exceeding climate awareness and, in some cases, levels of concern observed in wealthier countries. This concern appears to emerge less from formal knowledge of climate science than from the lived experience of local environmental changes. In addition, support for climate mitigation policies in developing countries is not lower than in high-income countries and, in many cases, is higher. These findings challenge assumptions that low awareness implies low demand, suggesting that the main barriers to climate action in many developing countries may lie not in public attitudes but in government responsiveness.

Awareness and Anthropogenic Causes

Climate awareness remains low in many parts of the developing world, especially sub-Saharan Africa and South and Southeast Asia. We draw on survey data from Afrobarometer and Latinobarómetro to explore climate awareness across countries within these regions (see **Figure 1**). In 2008, fewer than 40% of respondents in sub-Saharan Africa self-reported knowledge of climate change, with some countries reporting as low as 25–30%. As **Figure 1** shows, by 2023, mean climate awareness in Africa rose but was still only 51%. Comparatively, climate awareness is significantly higher in Latin America, at approximately 75% in 2010 and over 80% in recent surveys.

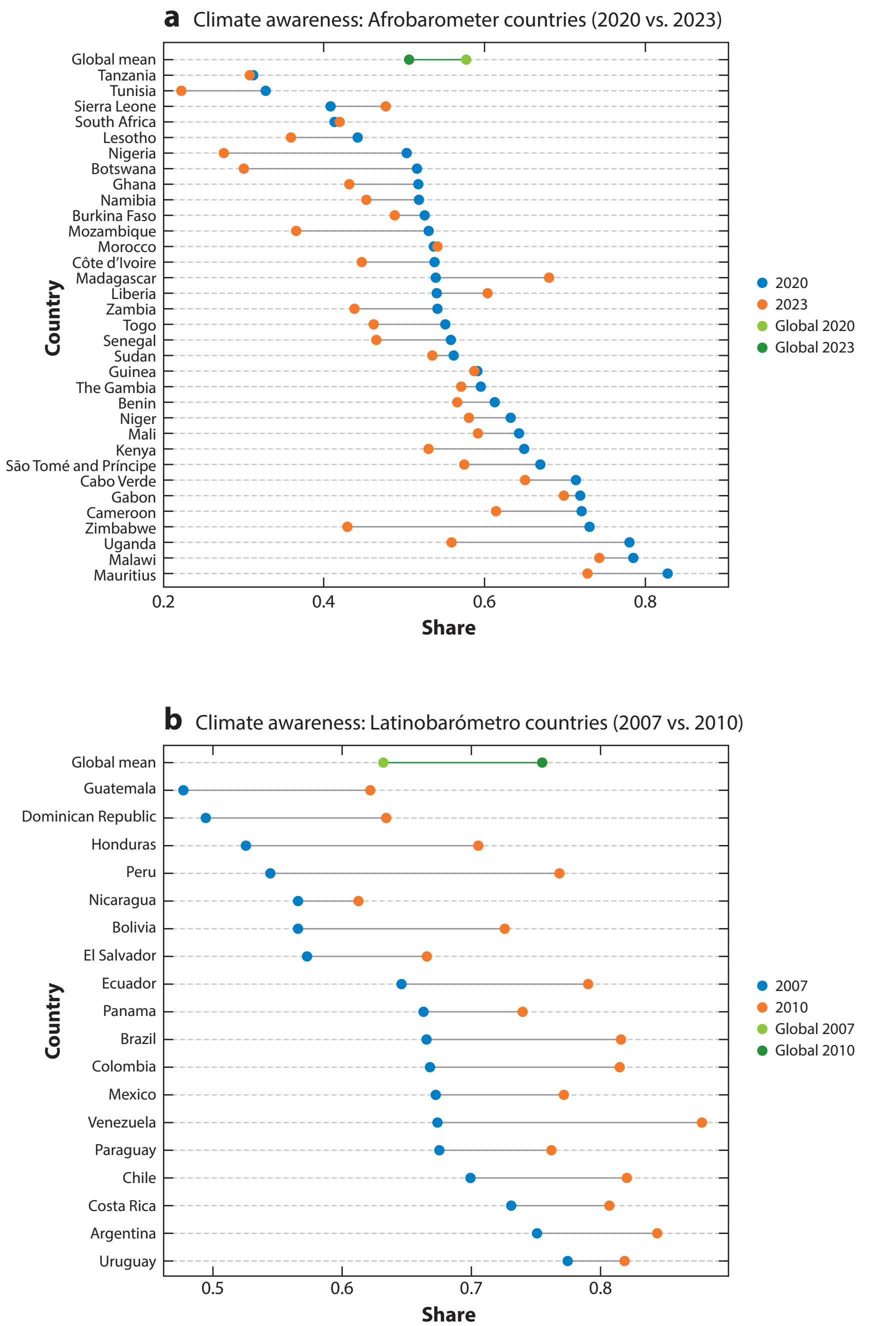


Figure 1

Climate awareness. This figure plots national (weighted) means of climate awareness. The wording of the Latinobarómetro (panel *b*) survey question is: "How much have you heard or read about global warming or climate change?" We recoded responses such that "none" and "a little" have the value of zero, and "some" and "a lot" have the value of one. The Afrobarometer (panel *a*) question is binary: "Have you heard about climate change, or haven't you had the chance to hear about this yet?" Data in panel *a* from Afrobarometer (2022, 2023). Data in panel *b* from Latinobarómetro (2007, 2010).

However, these levels still lag behind those reported for North America, Europe, and Japan, which report a mean climate awareness of over 90% (Lee et al. 2015).

Figure 1 reveals that climate awareness increased considerably in all Latin American countries between 2007 and 2010. In contrast, awareness declined in most African countries between 2020 and 2023. The study of attitudinal change over time remains a significant gap in the literature. Widespread data sparsity and inconsistencies in survey design—in particular, the lack of repeated questions across waves—make it challenging to study temporal trends in most developing countries. Overall, climate knowledge, attitudes, and their determinants remain poorly understood.

Low awareness does not reflect a lack of perceived change. Many populations, especially farmers, in LMICs report rising temperatures, soil toxicity, erratic rainfall, and water scarcity (Kabir et al. 2017). However, experiencing local environmental changes does not necessarily mean individuals can attribute these changes to an anthropogenic, global phenomenon. Consistent with this, in 2020, only 41% of Afrobarometer respondents in Africa identified human activity as the primary cause of climate change (Simon 2023).

Consistent with information-deficit models, access to climate information via education systems and the media is a central barrier to improved climate awareness in developing countries. Secondary schooling is strongly correlated with climate knowledge across most developing countries, including those in Africa (Simpson et al. 2021) and Latin America (Aklin et al. 2013, Spektor et al. 2023). Media access also drives climate awareness and shapes climate views (González & Sánchez 2022), especially in low-education contexts. Social media plays a growing but uneven role: Platforms such as YouTube and Instagram (but not Facebook) correlate positively with climate awareness (Gómez-Casillas & Gómez Márquez 2023). However, low media literacy can make social media users vulnerable to misinformation and climate change skepticism (Strudwicke & Grant 2020). The influence of digital information access on climate attitudes in developing countries deserves further study.

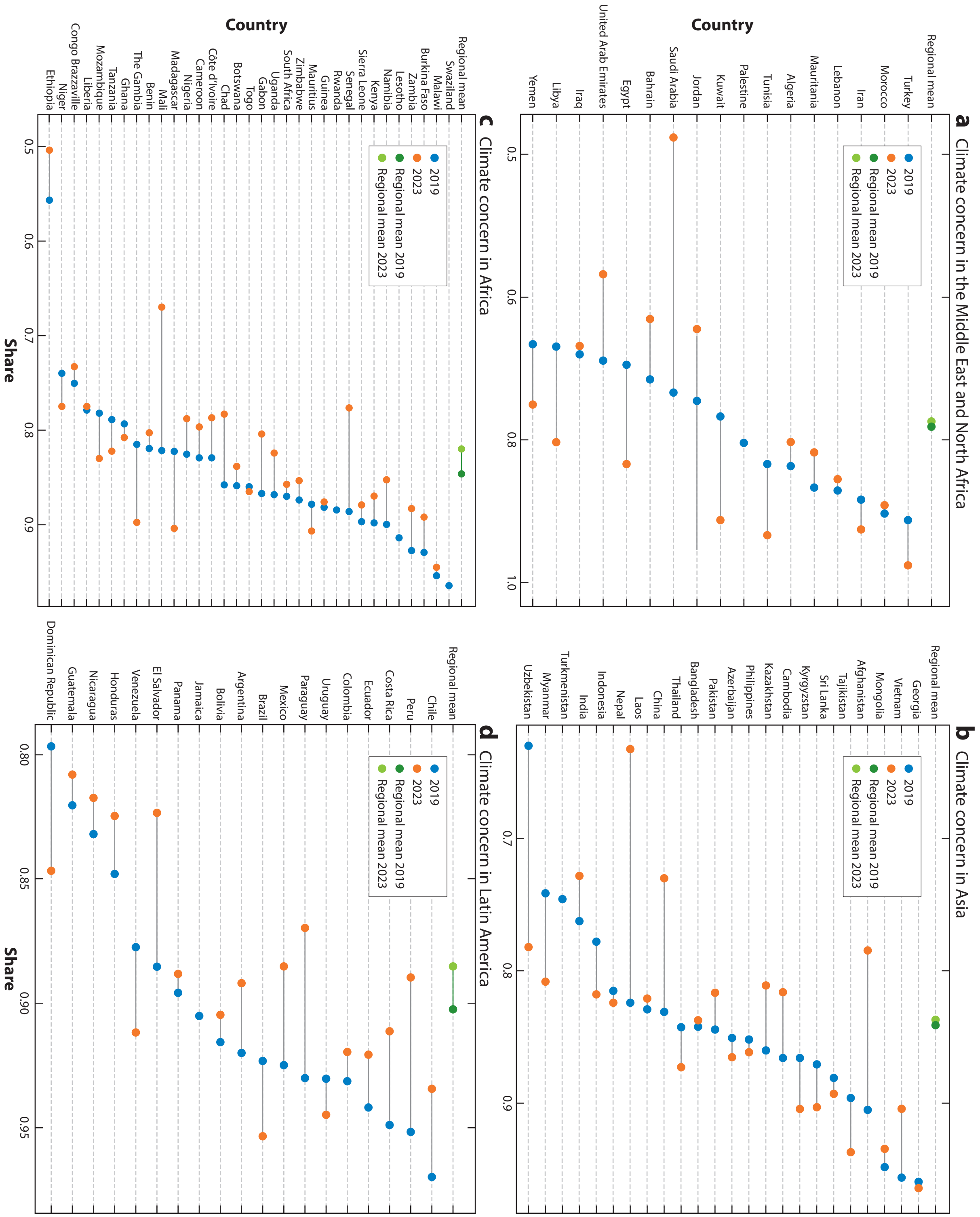
Climate Concern

We draw on survey data measuring climate concern levels in 2019 and 2024 from the Lloyd's Register Foundation's World Risk Poll for countries in all developing regions. As **Figure 2** shows, two patterns stand out. First, concern is high across the developing world, with rates of approximately 90% in Latin America, 83% in Africa and Asia, and 78% in the Middle East and North Africa—often matching or exceeding levels in high-income countries. Second, concern generally outpaces awareness in low-income countries. Contrary to some claims (e.g., Van der Linden 2015), being concerned about a changing climate does not require high climate awareness. Instead, local environmental changes, not abstract climate knowledge, are often the key drivers of concern.

Policy Support

Even when climate concern is high, addressing climate change through mitigation and adaptation policies does not directly become a top priority for the population. In the developing world, there is often the immediacy of other, more pressing issues. Measuring support for climate policies in developing countries is hindered by data scarcity and the sensitivity of responses to question wording. However, despite data limitations, existing evidence suggests relatively strong support for climate policies, even in the face of low awareness and limited state capacity.

The Gallup World Poll (2022–2023, 125 countries) includes an indirect measure of policy support: willingness to contribute 1% of household income monthly to combat global warming. In low-income countries, 60–70% of respondents expressed willingness (Andre et al. 2024). The Trust in Science and Science-Related Populism survey (2022–2023, 68 countries) directly



(Caption for **Figure 2** appears on following page)

Figure 2 (*Figure appears on preceding page*)

Climate concern. This figure plots national (weighted) means of climate concern in (*a*) the Middle East and North Africa, (*b*) Asia, (*c*) Africa, and (*d*) Latin America in 2019 (*blue*) and 2023 (*orange*) using World Risk Poll data (121 countries). The wording of the survey question is: "To what extent do you think climate change is a threat to your country in the next 20 years?" We recoded responses such that "not a threat at all" and "don't know" have the value of zero, and "somewhat serious threat" and "very serious threat" have the value of one. Data from Lloyd's Register Foundation (2019, 2024).

measures support for five climate policies. Analysis shows broad support in developing countries, particularly for expanding renewable energy and conservation measures, such as protecting forests (Cologna et al. 2025).

Support for different climate policies is not universal. Instead, support tends to increase when policies are framed in terms of environmental conservation, pollution, or agriculture—domains that overlap with, but are not identical to, climate policy. For instance, in China, climate-related topics rarely trend on social media platforms, such as Weibo, although air pollution once drove online backlash in the 2010s (Liu & Wallace 2023). Recent evidence shows that public complaints about environmental violations, when amplified through social media, can significantly reduce pollution by increasing the political salience of government enforcement (Buntaine et al. 2024). However, research from India shows that partisanship and sensitivity to personal costs can blunt public accountability, even for issues such as air pollution, whose health risks are widely recognized by citizens (Singh & Thachil 2023). Insofar as industrial polluters are also significant emitters of greenhouse gases, such mechanisms may have implications for climate outcomes as well.

Support for mitigation policies (e.g., carbon taxes or green infrastructure) increases when individuals perceive these policies to be fair (Dechezleprêtre et al. 2025). Recent work finds that mass support for international climate transfers increases when mitigation policies are framed around compensation (Gaikwad et al. 2022). Thus, an emerging strand of policy discourse repackages mitigation around "co-benefits," such as green jobs, technological innovation, and improved public health. However, a two-country survey experiment shows that these frames fail to attract more public support than the usual focus on reducing climate change risks (Bernauer & McGrath 2016). Variation in climate policy support is also a function of trust in government: It is higher when people view their governments as sufficiently competent (Meckling & Benkler 2024, Fesenfeld 2025). Conversely, low political trust, especially in contexts of perceived corruption or unresponsiveness, can dampen support even among those with high climate concern (Andrews et al. 2025).

Community attributes, such as (perceived) social norms and second-order beliefs, also explain variation in support. Where climate action is rare or stigmatized, people may hide their concern and policy preferences; where it is normalized among peers, concern and support for climate action rise (Cologna et al. 2025, Todorova et al. 2025). Climate attitudes spread through social signaling and cues. For instance, peers, including those on social media, shape one's perception of whether floods are natural hazards or attributable to climate change. Very few studies leverage these network dynamics, making this focus a promising avenue for future research.

Individual-level factors also shape climate policy support. People often oppose mitigation policies when they perceive personal costs to their way of life, and this perception increases with economic development and for men more than women (Bush & Clayton 2023). In contrast, those most vulnerable to climate impacts, including residents of coastal areas, small islands, or agricultural regions, are more likely to support climate action.[4] Again, vulnerability, not climate knowledge, is the primary driver of support. Unlike in high-income countries, partisan identity and ideology play a more minor role in developing countries. However, ideological proxies, such as

[4]Vulnerability itself becomes an identity that is more salient than ethnicity or class in mobilizing support (Eisenstadt & West 2019). However, Mildenberger et al. (2025) find that even among those most

individualism (Spektor et al. 2023) and belief in interventionist deities (González & Sánchez 2022), can reduce support for climate policy. These patterns align with political psychology theories (e.g., motivated reasoning), although such frameworks remain underused in this context.

To summarize, limited public investment in climate mitigation and adaptation in developing countries does not appear to reflect low public concern or widespread resistance to costly policies. One explanation is that climate policy lacks salience among voters; other issue areas take precedence in the developing world, although this possibility remains untested. In support of this view, Wappenhans et al. (2024) find that even after extreme weather events, increased public concern does not translate into greater political attention to climate issues (i.e., measured using political parties' public communication). Another possibility is that there is weaker government responsiveness and accountability in developing countries. For instance, Calacino et al. (2026) find that even when climate shocks prompt formal policy adoption, they do not necessarily lead to implementation, highlighting a persistent gap between symbolic policy adoption and actual change in policy outcomes. Measuring government response is itself difficult: Climate policy effort spans multiple stages—commitments, actions, and actual outcomes—each posing distinct challenges for observation and comparison (Lieberman & Ross 2025). Overall, these studies reveal a disconnect between climate impacts and political responsiveness, a key area for future research.

Way Forward

Public opinion research on climate attitudes in developing countries faces several limitations. First, despite a growing number of studies on the determinants of climate attitudes, the literature remains highly fragmented and undertheorized. The modal study regresses a measure of climate attitudes (e.g., climate awareness, belief in human causation, climate concern, or policy support) on numerous independent variables, before reporting which have the greatest predictive power (e.g., González & Sánchez 2022, Todorova et al. 2025). Such designs allow scholars to make progress in mapping the correlates of climate attitudes; however, a theory-driven portrait of the most important predictors across different types of climate attitudes remains elusive.

Second, data sparsity further limits public opinion research in developing countries. Standardized governance-focused national surveys often lack comprehensive questions related to climate change, reflecting low donor priorities for addressing climate issues. The Asian Barometer contains none; the Afrobarometer and Latinobarómetro ask about climate awareness in some rounds but omit questions on concern or policy support. Their temporal coverage is also low, hindering analysis of long-term trends in most developing countries. Inconsistent inclusion of climate questions across survey rounds further complicates efforts to track opinion over time. Limited topical depth further constrains efforts to link attitudes to behavior. Major climate opinion data sets (e.g., Yale Program on Climate Change Communication, World Risk Poll) offer better topical coverage but are limited in geographic scope. Furthermore, they are often designed by scholars residing in high-income countries, with less attention to locally salient framings, such as agricultural seasons.

Another important avenue for future work is testing different messaging, tools, and interventions to increase climate literacy (see Atkins et al. 2024). While this is an active research agenda in the Global North (e.g., Bergquist et al. 2020), there is a notable dearth of work on these topics in developing countries, particularly research sensitive to local conditions, concerns, and prevailing narratives and frames. From a policy perspective, research should focus on the conditions under which individuals and communities may prioritize climate-related investments

climate-vulnerable in the 55 small island states, there is wide variation in perceived climate risk and in beliefs about responsibility for resolving climatic threats.

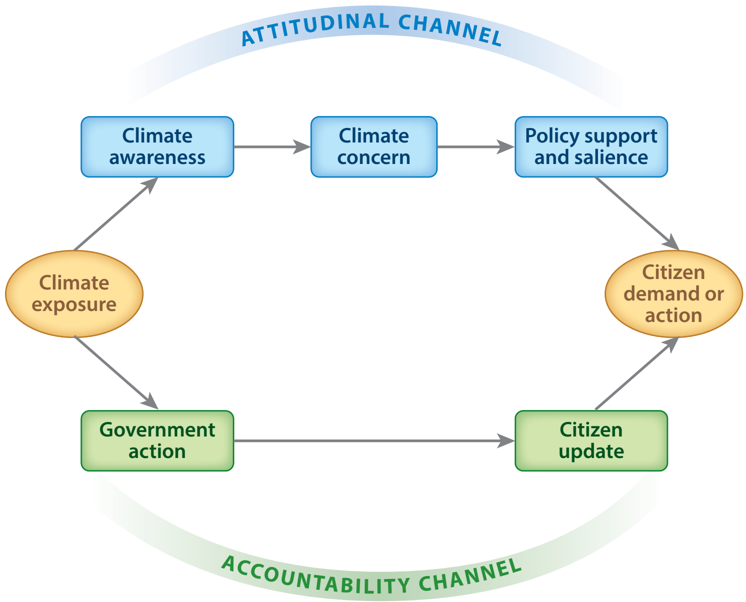


Figure 3

The accountability channel and the attitudinal channel are the two channels through which climate exposure is linked to citizen demand or action.

over other priorities, such as social protection, education, health care, and infrastructure, and the impact of climate literacy on supporting these investments.

THE EFFECTS OF CLIMATE CHANGE EXPOSURE

The previous section examined the determinants of climate attitudes in developing countries. A parallel and growing but largely disconnected literature investigates how exposure to extreme weather events, such as floods, droughts, and wildfires, shapes climate attitudes and behavior. As shown in **Figure 3**, we classify these studies into two strands that developed separately, rarely citing each other and relying on distinct theoretical frameworks. The figure illustrates the pathways linking climate exposure to citizen demand for more effective or responsive climate action. Much of the existing research isolates a single step in this causal chain (i.e., typically the effect of exposure on behavior, such as voting) without tracing the different intermediary legs of the chain. Our goal is to map the entire causal chain, identifying where the literature skips steps and where key gaps remain.

Attitudinal Channel

Researchers working within the attitudinal channel argue that climate inaction arises from the perceived psychological distance of its impacts—measured temporally or geographically (Keller et al. 2022)—and the uncertainty surrounding this distance (Sisco 2021). Direct, personal experience with climate risk is thought to reduce this distance and uncertainty, increasing emotional and cognitive engagement, climate concern, issue salience, and ultimately support for individual or collective climate action (Leiserowitz 2006). This effect may be especially pronounced in low-education contexts, where personal experience often resonates more than abstract information (González & Sánchez 2022). The effect may also be amplified when people face salient (i.e., flooding) as opposed to subtle (i.e., contamination of irrigation water with salinity due to rising sea levels) shocks (Patel 2025).

A central debate in this literature concerns how to conceptualize experiencing the consequences of climate change. Political scientists typically equate direct experience with objective exposure, measured through gridded climatic data sets. This approach risks ecological fallacies, especially when paired with highly aggregated survey data, and allows considerable researcher discretion in constructing exposure measures (Howe et al. 2019). Its advantage lies in treating exposure as exogenous to prior beliefs. By contrast, social psychologists emphasize subjective experience: For an event to shape attitudes, it must be perceived as unusual, personally relevant, and memorable. This perspective aligns more closely with theories of risk perception but introduces endogeneity, as those already concerned about climate change are more likely to interpret events as such. Unsurprisingly, self-reported exposure consistently predicts higher climate concern and policy support (González & Sánchez 2022, Spektor et al. 2023, Cologna et al. 2025, Dablander 2025), whereas objective exposure measures typically do not (Lee et al. 2015, Simpson et al. 2021, Xia et al. 2022, Cologna et al. 2025). Few studies, however, examine when—or whether—such updating in climate awareness, concern, or policy support translates into political outcomes, leaving a critical gap for future research.

Accountability Channel

In parallel to the attitudinal channel literature, where climate-exposed individuals supposedly support climate action after updating on climate risk, a different body of work explores the effect of (objective) climate exposure on electoral outcomes. In this literature, climate-exposed individuals use both a climatic event and the government's response to it (i.e., both its action and inaction) to update their beliefs about the government—in particular, about its capacity (Cole et al. 2012, Birch & i Coma 2023), responsiveness (Lazarev et al. 2014, Cooperman 2022), effectiveness in mobilizing resources (Blankenship et al. 2021), and trustworthiness (Ahmad & Younas 2021, Ahlerup et al. 2024). In some studies, individuals also update their capacity for collective action (e.g., Liu & Xu 2025, Balcazar & Kennard 2026). There is evidence that exposure to a climatic event can effectuate programmatic shifts toward left-wing and green party candidates who support redistribution and reconstruction (Visconti 2022, Pianta & Rettl 2025). Theoretically, updating is also a function of how the hazard is framed as part of political processes: by the press, opposition parties, friends, local leaders, or social media. However, we are not aware of a paper that rigorously explores the mediating factor of such narratives.

Accountability theories, particularly retrospective voting, primarily shape the link between climate hazards and voting behavior (Healy & Maholtra 2009, Achen & Bartels 2017, Ashworth et al. 2018, Rubin 2018, Baccini & Leemann 2020, Marsh 2025). Citizens use both the hazard's impacts and the government's responses to those impacts as a heuristic. If the government uses the event to signal responsiveness, it benefits electorally (e.g., Lazarev et al. 2014, Gallego 2018), but if it fails to respond adequately, it suffers electoral losses (Katz & Levin 2016). Government response is a function of various factors, such as electoral cycles (Cao et al. 2019, Cooperman 2022), incentives for promotion among local officials (Wu & Cao 2021), the scale of the climatic event (Birch & i Coma 2023), the alignment between local and national governments (Blankenship et al. 2021), and the logic of clientelistic exchange (P. Querubín & J. Labonne, unpublished manuscript).

Beyond voting, a nascent literature examines whether hazard exposure can directly influence government behavior and the structure of politics and institutions. Evidence is mixed regarding whether extreme weather events increase attention to environmental and climate issues among politicians and parties (Wappenhans et al. 2024). However, natural disasters shape who enters politics. In Brazil, floods decrease the average age and educational attainment of candidates, discouraging professionals with outside career options from running for office (Fasolin & Valentim 2024). In India, extreme temperature shocks before elections increase support for agriculture-oriented candidates campaigning on environmental issues (Amirapu et al. 2023). More fundamentally, disasters can also alter the institutional architecture of the state. Recent cross-national evidence suggests that natural disasters lead to increased fiscal and administrative centralization, particularly when events occur far from the capital or are geographically dispersed (Han et al. 2025). Disasters may encourage recentralization by highlighting the need for national coordination. These findings highlight that climate hazards may not only affect electoral behavior but also reshape the underlying distribution of authority within states. A vast literature also links climate shocks to conflict (for a review, see Koubi 2019).

Notably, in most studies, there is not anything unique about climate change per se. Scholars treat extreme climatic events just as any other exogenous shock (e.g., economic shock) that is sufficiently salient to voters to affect electoral outcomes (see Healy & Maholtra 2009, Achen & Bartels 2017, Ashworth et al. 2018, Baccini & Leemann 2020, Marsh 2025). This raises a central conceptual question: What, if anything, distinguishes climate shocks from broader economic shocks featured in the extensive literature on retrospective voting? In addition, another key puzzle remains: Climate exposure influences voting even when it does not affect climate attitudes or salience, a tension the literature has yet to resolve.

Thorny Methodological Problems

Both attitudinal and accountability studies fall short in providing theoretical clarity on the degree of exposure or prior climate awareness required to influence climate attitudes. What level or type of exposure shifts attitudes? Which hazards are most likely to shift attitudes: those most clearly attributable to climate change (e.g., heat waves), those most salient to the senses (e.g., floods), or those most destructive and thus most memorable (e.g., hurricanes and cyclones)? And is attitudinal change necessary for political or behavioral responses? These questions remain largely unresolved. These divergent conceptualizations of experience and hazards, in addition to the researcher's own degrees of freedom in how those are measured, create definitional and methodological tensions across studies (Keller et al. 2022).

The weak theoretical foundations of the social science literatures on extreme weather exposure, climate attitudes, and behavior are also reflected in the literatures' methodological inconsistencies. Researchers make arbitrary decisions about exposure metrics, relevant time scales (i.e., when the hazard event took place relative to the time of the survey or elections), and units of analysis (e.g., the grid, ADM2, ADM1, constituency, country) without clear theoretical justification (Howe et al. 2019). Both the attitudinal and accountability literatures also tend to focus on the reduced form with little attempt to unpack the different legs of the entire causal chain.

THE EFFECTS OF INSTITUTIONS ON CLIMATE MANAGEMENT

The preceding sections examined how citizens in developing countries perceive and respond to climate change. These analyses implicitly presuppose a political context in which public opinion matters and the populace can hold leaders accountable. However, across much of the developing world, political regimes are more varied, with many countries characterized by authoritarian or hybrid systems and limited electoral accountability. In these settings, public attitudes, while important, cannot fully explain climate outcomes. Political institutions operate differently in developing countries, and the pathways from citizen preferences to policy are often more indirect and complex.

This divergence foregrounds an important observation: Before examining how climate exposure generates political responses, we should step back to explore how institutions shape exposure in the first place. Climate change disproportionately affects developing countries; however, within

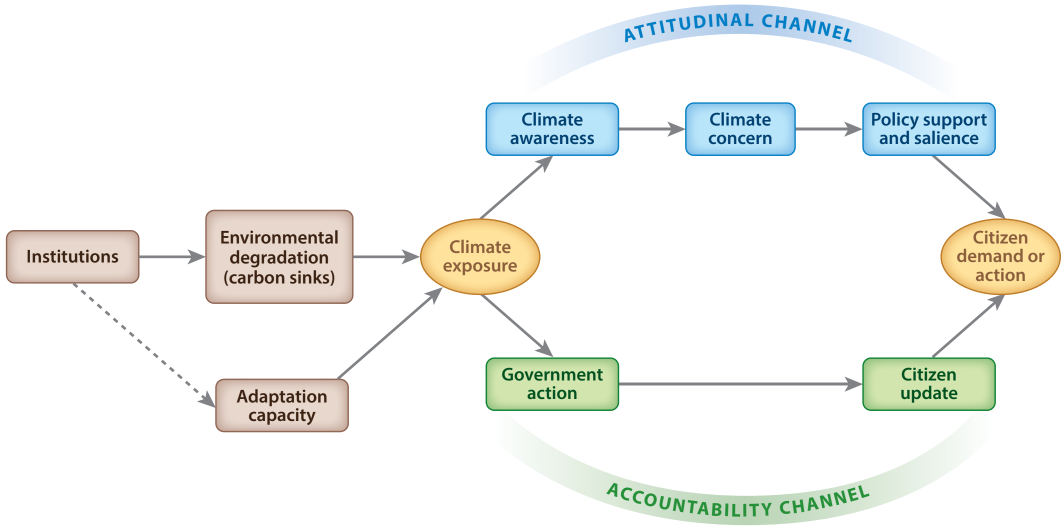


Figure 4

Extending the causal chain: institutions as upstream drivers of climate exposure and political response. Climate exposure is politically produced through the institutions that govern carbon sinks, build adaptive capacity, and allocate political voice.

the developing world, variations in effects are substantial. These differences in exposure are not simply a function of geography or poverty but rather are deeply rooted in national, subnational, and local institutions that structure both individual and collective responses. Institutional structures help explain why some governments protect carbon sinks while others allow their destruction, and why some communities develop resilience to climate shocks while others remain vulnerable. Climate vulnerability is not only inherited; it is also politically produced.

In this section, we examine how institutions structure the incidence, severity, and salience of climate exposure.[5] We extend the causal chain introduced earlier by identifying a precursor leg, as shown in **Figure 4**.

We treat climate exposure not as exogenous but rather as an outcome of political and institutional forces. Specifically, we analyze how political institutions at the national and subnational levels produce variation in climate vulnerability through three mechanisms: (*a*) governing the protection or destruction of carbon sinks, (*b*) shaping differential adaptive capacity, and (*c*) determining whose voice and authority matter over these decisions. These institutions are not merely background conditions that mediate climate effects; they are also political causes of climate exposure. We advance four core claims.

First, political institutions and sociopolitical attributes shape climate exposure and vulnerability. They precede it in the causal chain. Climate impacts do not simply happen to countries and communities but rather are politically produced. Differences in regime type, state capacity, and governance norms help explain why some jurisdictions raze carbon-rich forests or site factories in floodplains while others shield those assets. We lack systematic evidence on how institutions operate both as causal variables in generating differential vulnerability and as mediators conditioning the welfare effects of exposure to similar climate hazards. Cross-national data sets that link institutional design to granular exposure or impact data are rare.

[5]We refer to climate exposure to capture both climate exposure and climate vulnerability [see also technical definitions in the IPCC's Sixth Assessment Report (Pörtner et al. 2022)].

Second, because institutions sit upstream of climate vulnerability, they also shape the downstream political pathways reviewed in the section titled Climate Change Attitudes in the Developing World and in the section titled The Effects of Climate Change Exposure. In settings with weak state capacity, households and communities often rely on informal adaptation measures (e.g., migration, land repurposing, changing agricultural practices, livelihood diversification, water conservation, private cooling, or labor reorganization) to cope with climate change. These bottom-up responses may complement, substitute, or crowd out state action, muting the attitudinal and accountability pathways emphasized earlier. When informal strategies substitute for public provision, they can reduce pressure on governments and diminish demand for climate responsiveness. However, we know little about who adapts informally, when states step in to complement these responses, and how institutional incentives structure that balance.

Third, bridging the long-divided literatures on environmental and climate politics is essential for tracing causal chains. What was once framed as environmental politics—that is, the politics behind pollution control, deforestation, and land use—is often central to global climate governance. Environmental degradation is often the entry point through which individuals first experience climate change, making these local struggles central to any account of carbon production and climate awareness.

Finally, we highlight a paradox of representation: Those with the most granular knowledge of environmental change (e.g., Indigenous Peoples, forest-dependent communities, informal urban dwellers) are frequently excluded from arenas where adaptation policy and finance are negotiated. Although political science increasingly acknowledges this gap, systematic analyses of how political voice is allocated and its consequences for adaptation and resource distribution remain scarce (Gulzar et al. 2024, Clayton et al. 2025). The subsections that follow explore four key themes—carbon sink governance, adaptive capacity, political voice, and distributive justice—to demonstrate how institutions affect the response to climate change as well as its spatial and temporal dimensions.

Carbon Sinks

One of the most consequential ways political institutions shape climate change vulnerability is through the governance of carbon-absorbing ecosystems. This is especially critical in developing countries, which contain most of the world's remaining carbon sinks and biodiversity reserves—tropical forests, peatlands, and wetlands. These ecosystems serve as natural buffers against climate change, yet the literature often treats deforestation, pollution, contamination, and degradation as local environmental issues, obscuring their centrality to global climate dynamics. Reframing environmental degradation, especially deforestation, as the loss of carbon sinks reveals the shared terrain across the environmental and climate politics literatures. Yet few studies examine how such degradation translates into uneven and slow climate harms across space and communities.[6]

Regime type is an important dimension for understanding how governments manage ecosystems and balance economic development with environmental protection. Early work suggests that democracies perform better because of improved accountability and responsiveness (Midlarsky 1998, Li & Reuveny 2006). However, evidence from the developing world complicates this view. In their review of this literature, Baragwanath & Gulzar (2026) find that the correlation between democratic governance and environmental outcomes is, in fact, weak and inconsistent across contexts. Democratic competition can incentivize short-term resource extraction, particularly where state capacity is weak. In the Brazilian Amazon, for instance, Xu (2025) finds that

[6]For an exception, see Herrera (2024b).

competitive elections induce deforestation by incentivizing local politicians to weaken the bureaucratic capacity of environmental monitoring agencies. Sanford (2023) finds that democratic transitions correspond to an additional 0.8 percentage point loss in forest cover, as politicians trade forest concessions for electoral support.[7] Political competition can also entrench existing energy trajectories, reinforcing path-dependent reliance on fossil fuels unless green alternatives are already institutionalized (Aklin & Urpelainen 2013). In contrast, features of autocratic governments can allow for longer-term environmental planning by avoiding veto players and electoral volatility (Beeson 2010, Bayer et al. 2016, Clark et al. 2019, Beiser-McGrath et al. 2023).

Decentralization introduces another layer of institutional variation in carbon sink governance. Land use and environmental regulation delegated to local governments can enable policy innovation but also create opportunities for elite capture. In Argentina, governors weakened forest protections to avoid conflict with agribusiness (Milmanda & Garay 2019). Again in Argentina, subnational bureaucratic capacity explains differences in conservation outcomes (Alcañiz & Gutierrez 2020).[8] Political alignment with the federal government also increases the likelihood of local protected area designation (Mangonnet et al. 2022), and the role of nongovernmental organizations and activists cannot be understated (Hochstetler & Keck 2007; Anderson et al. 2019; Buntaine et al. 2021; E. Barham, E. Bayi & M.V. Murillo, unpublished manuscript).

At the local level, a literature mainly outside of political science documents Indigenous and local knowledge on carbon sink management and community-driven responses. Indigenous Peoples, comprising only around 6.2% of the global population, manage more than 25% of Earth's land surface and 40% of all terrestrial protected areas and ecologically intact landscapes (Garnett et al. 2018, ILO 2019). Their customary institutions play a critical role in mitigation and adaptation. In Southeast Asia, communities use local rules (e.g., logging bans, riverbank vegetation requirements, elevated storage) to manage flood risks (Hiwasaki et al. 2015). Despite this capacity, there is a more limited literature in political science on how states can support Indigenous Peoples with climate adaptation and mitigation without undermining their autonomy.

A rich literature examines Indigenous movements in Latin America, their demands for collective rights, and subsequent state recognition of land and self-governance (Yashar 2005, Van Cott 2010, Eisenstadt 2011). Recent work explores how collective recognition affects individual identity and decision-making within communities (Albertus 2025). Although not rooted in the climate or environmental politics literatures, this research illuminates how Indigenous rights shape the governance of land and natural resources that function as carbon sinks. With few exceptions (e.g., Martínez-Álvarez 2022, Baragwanath et al. 2023, Gulzar et al. 2024), studies rarely address legal recognition or co-governance models that protect rights while scaling successful environmental management practices. Strengthening Indigenous Peoples' and local communities' negotiating capacity offers another path to reform. In Liberia, which lost 15% of its tree cover between 2002 and 2024, communities often lease forests without fair compensation. Christensen et al. (2025) show that training these communities in interest-based negotiation both reduces deforestation and improves agreement value.

Payment for Ecosystem Services (PES) programs provide another pathway to incentivize conservation by compensating communities for their efforts. These programs aim to reduce practices such as logging and crop burning while also promoting poverty alleviation (Jayachandran 2023). However, data on the effectiveness of PES programs on deforestation, poverty, and other

[7] Calacino (2025) shows that the effects of electoral competition also hinge on the visibility of environmental harms to voters.

[8] Although their work does not focus on managing carbon sinks, Dipoppa & Gulzar (2024) similarly find that bureaucratic incentives and capacity have strong effects on patterns of air pollution from crop burning in South Asia.

outcomes are mixed. In Uganda, Jayachandran (2023) shows that payments reduced deforestation. In India, Jack et al. (2025) find that unconditional up-front payments increased compliance with crop residue burning. Yet a meta-analysis by Samii et al. (2014) concludes that PES programs have limited impacts. We refrain from reviewing in depth the vast literature on PES across other disciplines, a literature that tends to be technical in focus and rarely appears in mainstream political science outlets.[9] We note, however, that the political dynamics behind PES adoption and implementation are comparatively understudied. Large-scale rollout is challenging, as it requires integrating data on livelihoods, income, and land rights with institutional frameworks for payment delivery, particularly in remote, unbanked areas.[10] Future research could further investigate the political incentives and institutions, such as property rights, that shape the design and implementation of PES programs and whether similar models can be applied to marine and coastal conservation.

Adaptation to Climate Change

Political institutions further shape climate vulnerability by structuring adaptive capacity. Studies exploring differential vulnerability and voice in climate governance tend to build on the environmental justice framework. Developed initially around struggles over toxic waste siting, environmental justice scholarship traces how environmental harms and access to public protections map onto race, class, gender, and other social divides (Walker 2012). Climate scholarship extends these insights, showing that sea level rise, drought, and heat waves disproportionately impact lower-income countries and marginalized communities (Dolšak & Prakash 2022). We treat environmental justice as encompassing both environmental and climate justice and emphasize the institutional forces that shape these inequalities.

The emerging literature on adaptation politics sheds light on how sociopolitical dynamics structure both top-down and bottom-up responses. This literature documents how migration (Arias & Blair 2022, Draper 2022) and labor formalization (Liu & Xu 2025) are strategies for climate adaptation.[11] Dependence on migration (Vinke et al. 2020) and informal community adaptation strategies can absolve governments of responsibility for long-term in situ adaptive planning. Recent work demonstrates that while such private investments can cushion heat and income losses, they exacerbate existing inequalities (Carleton et al. 2024). Because these bottom-up strategies often substitute for drainage upgrades, social protection, or early warning systems, they can blunt pressure for state-led programs, weakening the attitudinal and accountability channels outlined earlier in this article. However, we lack systematic, comparative understanding of the institutions and political structures that determine this substitution logic—when and why bottom-up adaptation crowds out, complements, or catalyzes government action.

Emerging research underscores the unintended consequences of bottom-up adaptation efforts. Designed to reduce vulnerability, such initiatives can backfire when poorly planned or politically captured, a process known as maladaptation (Dolšak & Prakash 2018). Adaptation programs should therefore be evaluated not only for technical efficacy but also for their distributional effects (Eriksen et al. 2020). Far from being politically neutral, adaptation is shaped by power relations

[9]This article does not address the literature that falls mostly outside of political science on Reducing Emissions from Deforestation and forest Degradation in developing countries (REDD+). Under this framework, developing countries can receive results-based payments for emission reductions when they reduce deforestation.

[10]For an assessment of the impact of different PES schemes on livelihoods in contexts with varying land tenure arrangements, see Mahanty et al. (2013).

[11]This article refrains from surveying the vast literature on climate and migration. We limit the focus here to noting migration as form of adaptation.

that can reinforce existing inequalities. Once implemented, it also carries political implications. Studying a water scarcity program in Mexico, Calacino & Martinez-Álvarez (2025) find that voters reward politicians for adaptation efforts, suggesting that these efforts may be less contentious than mitigation.[12]

Top-down adaptation programs, too, often fail when they ignore local realities or political dynamics. Climate aid frequently follows donor priorities rather than local needs (Gaikwad et al. 2025). Protective infrastructure and relocation schemes may displace the poor (Sovacool & Linnér 2016), and conservation programs sometimes exacerbate environmental degradation. In Aceh, Indonesia, a youth ranger program improved economic outcomes and modestly decreased illegal logging but was associated with increased small-scale mining (Paler et al. 2015). These examples highlight that adaptation and mitigation programs must be both technically sound and locally grounded to avoid unintended harm.

Despite these challenges, national and subnational governments are increasingly investing in top-down climate adaptation programs that integrate national policy with local implementation. Indonesia's ProKlim program supports village-level mitigation and adaptation. Brazil's AdaptaCidades integrates resilience planning across 11 states. Kenya's County Climate Change Funds empower local communities to manage climate finance (Crick et al. 2019), and Nepal and the Philippines have institutionalized local adaptation priorities through national frameworks (Woodruff & Regan 2019). Social insurance programs can buffer climate-induced income loss, yet the institutional conditions shaping these programs' effectiveness remain poorly understood.

Successful adaptation depends not only on program design but also on the integration of local knowledge. As discussed, Indigenous Peoples and local communities, although socioeconomically marginalized, often possess deep insight into environmental variability (Ramos-Castillo et al. 2017). A global review of 119 studies identified 1,851 locally led adaptation responses (Schlingmann et al. 2021). A recent randomized controlled trial in Indonesia found that combining tailored climate information with deliberative processes increased support for local climate projects (Erbaugh et al. 2025). These findings highlight the importance of participatory institutions that incorporate community-specific knowledge.

However, significant knowledge gaps remain. Case studies abound, but cross-contextual analysis of local adaptation strategies—by states, communities, or households—is rare, mainly due to data limitations. Similarly, public opinion research has largely overlooked demand for adaptation policies in developing country contexts. While public support for mitigation is increasingly studied, we know little about how individuals evaluate adaptation policies, which groups demand them, and how they prioritize them relative to other urgent needs under fiscal constraints. This is, in part, a result of how climate adaptation is not as well defined as mitigation. Existing research often conflates climate impacts with general policy salience, without distinguishing between support for adaptation and mitigation (Andre et al. 2024, Hornsey & Pearson 2024). Much of the existing adaptation literature focuses on household- or firm-level behavior and is grounded primarily in economics (for a review, see Carleton et al. 2024), with less attention to community-level dynamics or the political economy factors (e.g., social networks, local institutions, governance arrangements) that shape either top-down government investments or bottom-up collective adaptation.

Where comparative work exists, it tends to focus on the role of institutions in shaping climate aid distribution (see the work by Michaelowa & Michaelowa 2012, Falzon 2023). International green aid often favors countries with institutional credibility or geopolitical alignment over those with the greatest need (Gaikwad et al. 2025). Subnational capacity is also consequential: Some local

[12]This may reflect that many adaptation initiatives, such as irrigation in rain-fed regions or clean cookstove programs, are welfare-enhancing development projects.

agencies enact protective policies, while others, constrained by limited resources and discretion, see climate shocks deepen existing vulnerabilities. These gaps are not just technical. They are political choices about investment, voice, and inclusion.

Climate Politics as Distributive Politics

Political institutions further shape climate vulnerability by structuring who participates in environmental management and climate governance. Inclusion determines whether conservation or adaptation policies succeed, how to distribute benefits and burdens, and how these distributions might exacerbate existing inequalities. Early international relations scholarship framed climate change as a global commons problem—a tragedy of the commons requiring sovereign cooperation (e.g., Barrett 2003). This logic remains central to research on emissions reduction and international cooperation. However, as climate impacts become more visible and spatially uneven, they are increasingly understood as distributive conflicts—over costs, risks, and political representation (Roberts & Parks 2006, Aklin & Mildenberger 2020, Colgan et al. 2021, Alcañiz & Gutiérrez 2022). In a recent review article, Ross (2025) extends this view by tracing how distributive conflicts over climate policy play out across sectors, social groups, and levels of governance, emphasizing the concentrated veto power of reform losers.

This distributive turn reframes both mitigation and adaptation. Decarbonization generates global public goods but imposes localized costs. For example, in fossil fuel–dependent communities, achieving equitable green transitions is a political necessity. The literature on just transition policies illustrates these distributive tensions. Decarbonization policies often generate backlash from workers and communities whose livelihoods depend on fossil fuel industries, creating political challenges that many governments struggle to manage effectively. Comparing Brazil and South Africa, Hochstetler (2020) shows that energy transitions are shaped not only by international pressure or environmental need but also by the political coalitions, institutions, and development models that structure state–market relations. A global analysis of 32 fossil fuel reforms, such as raising gasoline taxes and reducing fuel subsidies, reveals that most are reversed within 5 years (Martinez-Alvarez et al. 2022). Similarly, Mahdavi et al. (2022) demonstrate that between 2003 and 2015, net fossil fuel taxes and subsidies remained essentially unchanged, with policy stasis driven more by fiscal constraints than political opposition. In response to these challenges, governments are experimenting with compensatory policies designed to support affected populations through job retraining programs, infrastructure investment, green economic development, and redistribution of carbon tax revenues (Gaikwad et al. 2022). However, although politically necessary, compensation rarely occurs at the scale or credibility needed to sustain reforms (Ross 2025). The credibility of these promises is central, as communities often fear policy reversal or neglect (Gazmararian & Tingley 2023). Evaluating the effectiveness of such compensatory programs represents a promising avenue for future research. Inclusive, well-designed transitions can be practical and electorally rewarding.

Adaptation, too, is fundamentally distributive: It involves allocating public goods and infrastructure in the face of unequal vulnerability. Adaptation is not just a technocratic necessity but also a political struggle over resources and representation. The existing literature identifies several institutions that support adaptation and the management of carbon sinks by marginalized communities. As discussed, programs that enhance land tenure and local authority, especially among Indigenous Peoples, have been linked to reduced deforestation (Martínez-Álvarez 2022, Baragwanath et al. 2023, Gulzar et al. 2024). Participatory mechanisms can further enhance targeting and legitimacy, particularly when supported by robust enforcement capacity and adequate funding. Community monitoring can strengthen compliance with deforestation limits (Slough et al.

2021) and wetland management (Herrera 2024a). By contrast, participation without institutional support often fails to achieve its goals. In Ecuador, land titling and participatory reforms had little effect on deforestation in the absence of state backing (Buntaine et al. 2015). Inclusion alone is insufficient; effective policy must link participation to power and adequate resources.

Institutional design also mediates how social and ethnic cleavages shape climate outcomes. Ruling coalitions may shield co-ethnics from environmental risks (Dawson et al. 2025), and support for climate policy varies by identity and proximity to political power (Zucker 2022). Intergroup contact between climate-vulnerable in-groups and out-groups can increase support for inclusive climate action (Gaikwad & Zucker 2024), but institutional channels are necessary for those preferences to translate into sustainable policy.

In summary, across the developing world, political institutions significantly influence the implementation and inclusivity of climate policy, the preservation of carbon sinks, the scope of adaptation efforts, and the equity and legitimacy of policies. These foundations are crucial for understanding the political roots of climate exposure and designing effective climate governance. Three cross-cutting insights emerge. First, treating pollution, deforestation, or land use as exclusively environmental rather than also climate politics obscures their role in managing the planet's carbon sinks. Bridging these literatures reveals that classic political science concerns, such as state capacity, clientelism, power inequality, and regulatory capture, remain central to climate outcomes. Second, adaptation is not a technocratic add-on; it is a distributive arena shaped by institutional gaps. Bottom-up responses may cushion shocks, but they can also have negative unintended consequences, and they may substitute for state provision, thereby dampening demand for public action. Third, those with the most place-based knowledge—Indigenous Peoples, local communities, forest dwellers, and informal residents—are often excluded from the arenas where climate finance and rules are made, perpetuating a deep representation gap.

Despite a growing literature, major blind spots remain:

1. **Institutional drivers of exposure**. We still lack systematic evidence on how different political institutions shape exposure and mediate climate impacts. Data sets linking institutional design to fine-grained hazard and outcome data across contexts are rare.
2. **The politics of adaptation**. Research has only begun to explain when bottom-up strategies crowd out, complement, or catalyze public programs—and how fiscal or electoral incentives shape that balance. Comparative studies on institutional conditions and public preferences for adaptation are urgently needed.
3. **Representation and justice**. We know little about how different institutions include in or exclude from climate decisions Indigenous Peoples, local communities, and vulnerable groups, or how that exclusion affects exposure and legitimacy. Political voice should be treated not only as a normative good but also as a variable shaping exposure and resilience.

Addressing these gaps requires theory-driven measures of institutional design and better data on local adaptation and finance flows. The challenge now is not only to document exposure but also to explain—and ultimately redress—the political processes that produce it.

CONCLUSION: NEW DIRECTIONS FOR A CHANGING CLIMATE

Climate change disproportionately affects the developing world and is projected to push between 32 and 132 million people in these regions into extreme poverty by 2030 (Jafino et al. 2020). In recent years, we have made important progress in understanding the politics of mitigation, especially in high-income democracies. Despite growing attention to climate governance, its institutional and distributive dimensions in the developing world remain critically understudied.

The current literature on climate change remains fragmented, often focusing on isolated outcomes or assuming reduced-form relationships without addressing underlying mechanisms. This article calls for deeper political science engagement with climate challenges in developing regions. We also emphasize the need to understand climate change through a comprehensive causal chain, from environmental shocks to individual awareness, policy preferences, issue salience, and ultimately political and social behavior—and in reverse, how politics shapes risk exposure in the first place. Several core insights emerge from our review. First, contrary to assumptions, concern about climate change is often higher in developing countries, despite lower levels of formal climate knowledge. Second, climate exposure does not consistently translate into shifts in political behavior. Its effects depend on prior beliefs, local context, and whether institutional channels enable interpretation and response. Third, we emphasize the role of institutions in structuring the distribution of climate risk and highlight the importance of integrating local knowledge and participatory mechanisms into climate governance.

These findings point to a research agenda that treats climate vulnerability as a political outcome and climate adaptation as also a deeply political process shaped by exclusion, inequality, and struggles over representation in climate decision-making. Future work should investigate how institutions shape exposure and response, how individuals understand and act on risk, and how distributive conflict plays out through adaptation. As climate finance expands, bridging these knowledge gaps is urgent—not only to inform policy but also to ensure that the most vulnerable communities are empowered to shape and benefit from climate solutions.

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